

CPE 486/586: Deep Learning for Engineering Applications

11 Graph Neural Network

Spring 2026

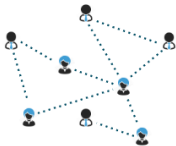
Rahul Bhadani

Outline

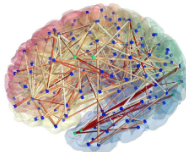
1. Motivation

Motivation

Graph/Network Dataset



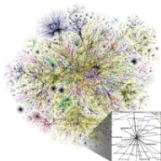
Social networks



Biological networks



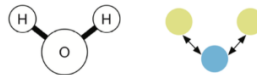
Transportation networks



Communication networks



n-body system

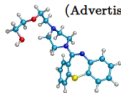


molecule

Graph/Network Dataset



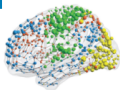
Social networks
(Advertisement)



Drug/Material
molecules
(Chemistry)



3D Meshes
(Computer
Graphics)



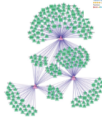
Brain
connectivity
(Neuroscience)



Transportation
networks



Word relationships
(NLP)



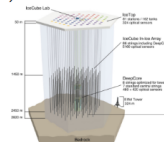
Gene Regulatory
Network



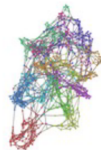
Scene understanding



Recommender
systems (Amazon,
Netflix)



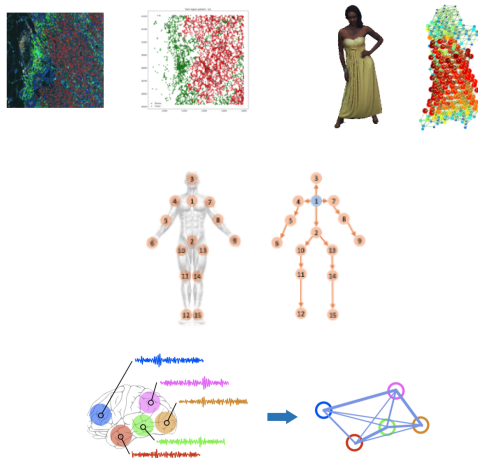
Neutrino
detection (High-
energy Physics)



Graph

Graphs to Represent Dataset

- ⚡ Graphs can be used to represent relationships in a structured data.
- ⚡ They provide a different perspective of dataset.
- ⚡ Impose relational inductive bias in data
- ⚡ Lead to knowledge discovery



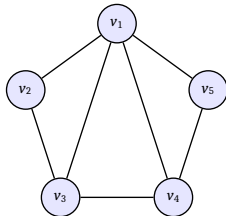
What is Graph?

A graph G is a data structure described by a set of edges and vertices (also known as nodes):

$$G = (V, E) \quad \text{or} \quad G = (V, E, A)$$

- ⚡ V set of vertices.
- ⚡ E set of edges.
- ⚡ They can be directed or undirected.

A graph is often represented by an Adjacency matrix, A . If a graph has N nodes, then A has a dimension of $(N \times N)$. People sometimes provide another feature matrix to describe the nodes in the graph. If each node has F numbers of features, then the feature matrix X has a dimension of $(N \times F)$.



What is Graph?

Graph features:

- ⚡ Node features: h_i, h_j (e.g. Atom Type)
- ⚡ Edge features: e_{ij} (e.g. bond type, time to travel from one traffic intersection to another, etc.)
- ⚡ Overall graph features: g (e.g. molecular energy)

Graphs are non-Euclidean Data

Suitable for data that resides in irregular, non-grid-like structures where relationships are not confined to regular Euclidean spaces.

- ⚡ Manifolds: Curved surfaces, e.g., 3D shapes or mesh data.
- ⚡ Point Clouds: Sets of points in 3D space without a grid structure (e.g., LiDAR data).

Real-World Data is Often Non-Euclidean.

Challenges in handling Non-Euclidean Data

⚡ fixed Grid Assumptions in MLPs, CNNs, RNNs

- Assume regular, structured data (e.g., grids or sequences).
- Cannot directly handle irregular neighborhoods or variable node connectivity in graphs or other non-Euclidean structures.

⚡ Irregular Neighborhoods:

- Varying numbers of neighbors per node.
- No uniform notion of proximity or direction.
- Standard convolution filters (which operate on fixed local neighborhoods) fail to adapt to these variable structures.

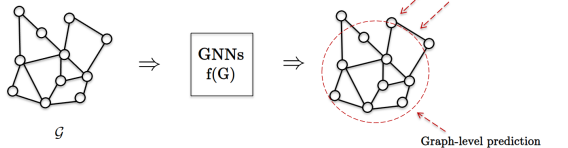
⚡ Lack of Spatial Regularity:

- The concept of "locality" is not fixed and varies across the structure.
- Non-Euclidean data like point clouds, graphs (undirected) is unordered.

We need a permutation invariance / equivariance.

Learning Tasks on Graph-structured Data

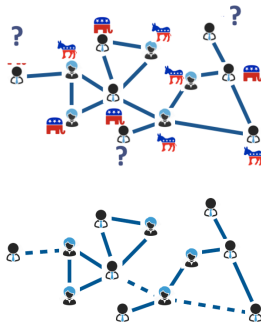
- ⚡ Graph-wise classification
- ⚡ Vertex-wise classification/inference of missing values
- ⚡ Link prediction



is it a 2?
is it a 4?



condition?
no condition?

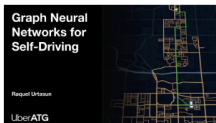
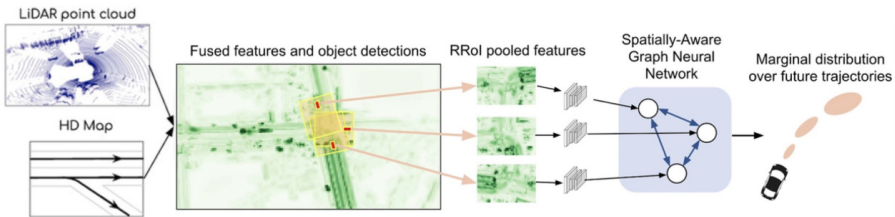


GNN Case Studies

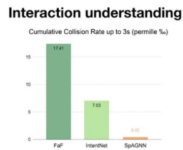
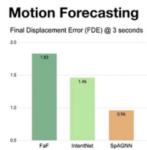
- ⚡ Chip design (Google)
- ⚡ Resource management
- ⚡ Scene reasoning (Meta)
- ⚡ Recommendation (UberEats/ Pinterest/ Alibaba)
- ⚡ Fake news detection
- ⚡ Finance
- ⚡ Natural Language Processing
- ⚡ Knowledge graphs (Amazon)
- ⚡ Transportation (Google Map, Waymo, etc.)
- ⚡ Autonomous driving (NVIDIA)
- ⚡ Protein & drug design (Google DeepMind/ Microsoft/ AstraZeneca)
- ⚡ Energy physics & simulations (Google DeepMind)
- ⚡ Code bug detection
- ⚡ Genomics

GNNs for autonomous driving

<https://slideslive.com/38930570/graph-neural-networks-for-selfdriving>

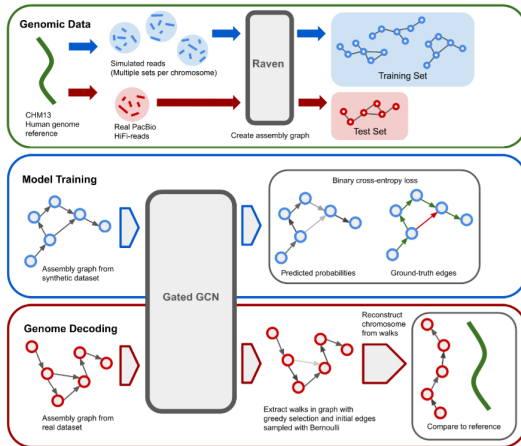


<https://slideslive.com/38930570/graph-neural-networks-for-selfdriving>



GNNs for Genomics

<https://arxiv.org/pdf/2206.00668>



Recommended Textbooks

- ⚡ Graph Representation Learning Book, Springer, 2020
https://github.com/RHxW/CV-DL-Docs/blob/master/GRL_Book.pdf
- ⚡ Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges, 2021
<https://arxiv.org/pdf/2104.13478>
- ⚡ Graph Neural Networks: Foundations, Frontiers, Applications, Springer, 2022
<https://graph-neural-networks.github.io/>

GNN Library in PyTorch

⚡ PyTorch Geometric

⚡ Released in March 2019

⚡ <https://pytorch-geometric.readthedocs.io/en/latest/>

Graph Categories: Natural Graphs

Graphs that are not artificially hand-crafted/constructed but just occur by natural process.

- ⚡ Social networks : Meta, LinkedIn, Twitter, Tiktok
- ⚡ Biological networks : Brain connectivity & functionality, gene regulatory networks
- ⚡ Communication networks : Internet, networking devices
- ⚡ Transportation networks : Trains, cars, airplanes, pedestrians
- ⚡ Power networks : Electricity, water



Minnesota Road Network



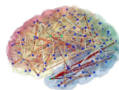
Telecommunication Network



US Electrical Network



Facebook



Brain Connectivity

=



Graphs

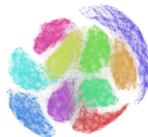
Graph Categories: Explicit Graph Construction from Data

Graphs constructed from data :

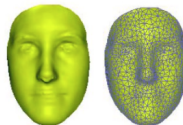
- ⚡ An example can be: compute the correlation between samples and treating samples as nodes, form an edge if correlation between two sample is greater than some threshold.



GTZAN Music Graph



MNIST Image Graph



3D mesh points

Adjacency Matrix

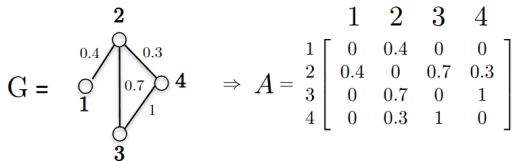
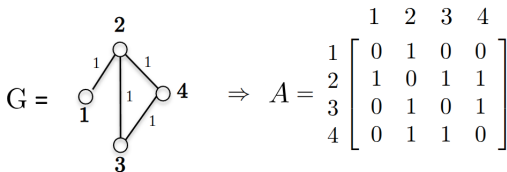
⚡ Matrix A in $G = (V, E, A)$ represents structural/topological information about the network. We have two classes of A .

- Binary matrix: $A_{ij} \in \{0, 1\}$
and

$$A_{ij} = \begin{cases} 1 & \text{if } (i,j) \in E \\ 0 & \text{otherwise} \end{cases}$$

- Weighted matrix: $A_{ij} \in [0, 1]$:
commonly normalized to

$$A_{ij} = \begin{cases} \in (0, 1] & \text{if } (i,j) \in E \\ 0 & \text{otherwise} \end{cases}$$



Curse of Dimensionality

In high dimensions, Euclidean distance between data becomes meaningless.

If data are uniformly distributed in $\mathbb{R}^d$, then pick any data point $\mathbf{x}_i \in V$, that gives

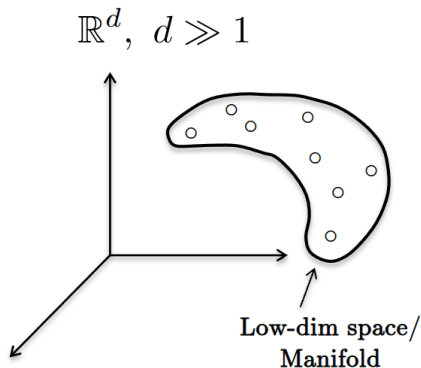
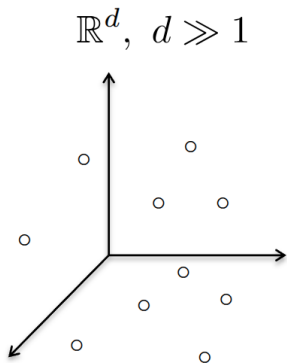
$$\lim_{d \rightarrow \infty} \mathbb{E}_{\mathbf{x}_i} \left(\frac{d_{\max}^{\ell_2}(\mathbf{x}_i, V / \mathbf{x}_i) - d_{\min}^{\ell_2}(\mathbf{x}_i, V / \mathbf{x}_i)}{d_{\min}^{\ell_2}(\mathbf{x}_i, V / \mathbf{x}_i)} \right) = 0$$

which just means that all data are really far away to each other with the same distance in a very high-dimensional space.

- ⚡ As an example, there is a loss of intuition in high-dimension.
- ⚡ You can think that in Normal distribution, in 1-d, most data are concentrated at the center.
- ⚡ But in high-dimensional spaces, most data are concentrated on the surface.

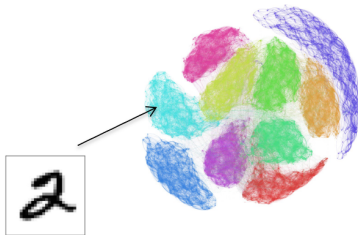
But Then There Might Be Some Structure in High-Dimension

- ⚡ **Data are uniformly distributed** is actually not true for real-world data.
- ⚡ Data have always properties, i.e. structures or invariances, such that they belong to a low-dimensional space called manifold where (geodesic) distances are meaningful.

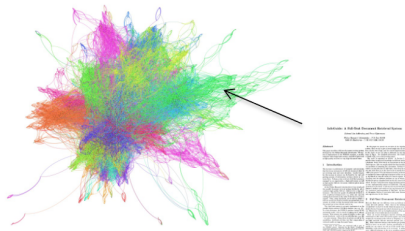


Manifold Learning

A class of algorithms that extracts low-dimensional patterns is manifold learning.



MNIST Image
Graph

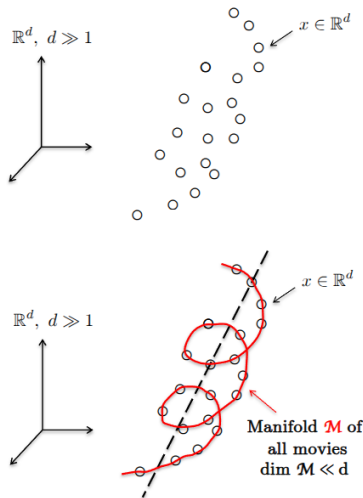


Graph of Text Documents
20newsgroups

Manifolds to Graphs

Primary assumption: High-dimensional data are sampled from a low-dimensional manifold.

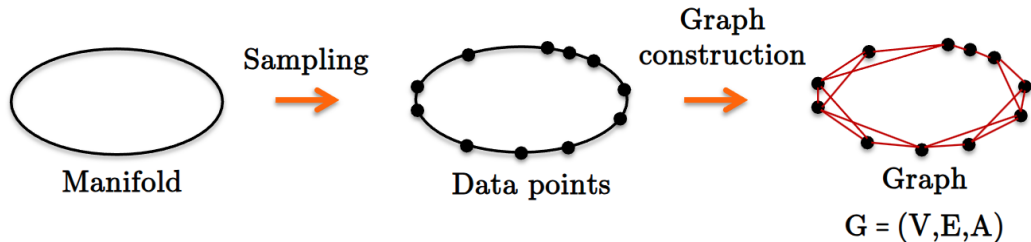
Example: Let x be a movie, each movie is defined by d features/attributes like genre, actors, release year, origin country, etc such that $\mathbf{x} \in \mathbb{R}^d$. We can decide to make the assumption that all movies form a manifold $\mathcal{M}$ in $\mathbb{R}^d$.



From Manifolds to Graphs

Graphs can be regarded as a manifold sampling process.

- ⚡ The manifold information is represented by a neighborhood graph (observe that the manifold is never directly observed or constructed).

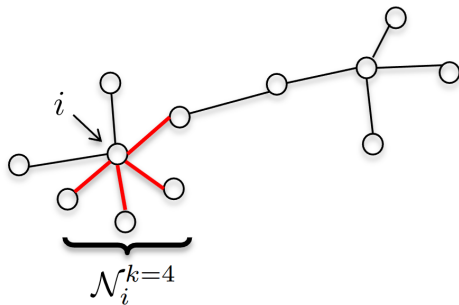


Neighborhood Graphs

⚡ Most populars are $k - NN$ graphs defined as :

$$A_{ij} = \begin{cases} e^{-\frac{dis(\mathbf{x}_i, \mathbf{x}_j)^2}{\sigma^2}} & \text{if } j \in \mathcal{N}_i^k \\ 0 & \text{otherwise} \end{cases}$$

where $dist(\mathbf{x}_t, \mathbf{x}_j)$ can be any kind of distance. $\mathcal{N}_i^k$ is the neighborhood of data $\mathbf{x}_i$.



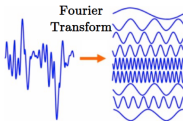
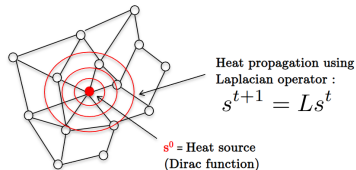
Spectral Graph Theory

Spectral graph theory (SGT) can

- ⚡ find meaningful patterns that reveal multi-scale graph structures.
- ⚡ Process data defined on the graph domain (a.k.a. graph signal processing).
- ⚡ Boost performance of learning tasks s.a. clustering, classification, recommendation, etc.

Graph Laplacian Operator L .

- ⚡ Central operator of diffusion processes on graphs.
- ⚡ Basis functions of this operator are the well-known Fourier modes



Graph Laplacian

⚡ Unnormalized Graph Laplacian:

$$L = D - A$$

where D is the degree matrix : $D = \text{diag}(d_1, \dots, d_n)$, and $d_i = \sum_j A_{ij}$.

⚡ Normalized Laplacian

$$L_n = D^{-1/2} L D^{-1/2} = I_n - D^{-1/2} A D^{-1/2}$$

⚡ Random Walk Laplacian:

$$L_w = D^{-1} L = I_n - D^{-1} A$$

⚡ All Laplacians are diffusion operators.